

FINAL PROJECT

EXPLAINING POINTS PER GAME IN THE NBA USING
LINEAR REGRESSION

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MACHINE LEARNING

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Introduction and research question

Scoring is one of the most important aspects of performance in professional basketball. Points per game are often used as a direct indicator of a player's offensive contribution, but this outcome can depend on several underlying factors such as playing time, shot selection, and shooting efficiency.

The objective of this project is to analyze which offensive statistics best explain the number of points scored per game by NBA players during the 2024/25 regular season. To answer this question, a multiple linear regression model is estimated using real player level data.

The main research question of this study is: Which individual offensive statistics are most strongly associated with points per game in the NBA?

Data and sample description

The data used in this analysis were obtained from the official NBA statistics website (nba.com), specifically from the Traditional and General Players statistics section in per game format for the 2024/25 regular season. The dataset contains individual statistics for NBA players, including scoring, shooting, and playmaking variables.

After cleaning the data and removing players with zero minutes played, the final sample consists of 519 players. Given the number of observations relative to the number of explanatory variables, the sample size is considered sufficient for estimating a multiple regression model.

PLAYERS		TEAM	AGE	GP	W	L	HR	PTS	FGM	FGA	FTM	FTA	FTOL	ORB	DRB	REB	AST	TOW	STL	BLK	PF	FOU	TSO	OW	
1	Alvin Robertson	IND	20	70	63	10	1	108	142	375	112	248	519	21	57	375	79	88	98	24	12	12	541	61	00
2	George Gervin	SD	20	70	40	27	20	104	118	322	65	121	213	65	108	317	121	32	113	5	21	12	574	56	11
3	Walt Frazier	SEA	20	70	40	26	10	107	296	712	263	576	120	62	417	52	64	100	10	33	22	643	59	340	85
4	LaRue Harris	SEA	20	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
5	George Johnson	SEA	20	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
6	James Jones	SEA	27	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
7	John Lucas	SEA	27	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
8	James Tate	SEA	20	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
9	James Tate	SEA	20	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
10	James Tate	SEA	20	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
11	James Tate	SEA	20	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
12	James Tate	SEA	20	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
13	James Tate	SEA	20	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
14	James Tate	SEA	20	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
15	James Tate	SEA	20	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
16	James Tate	SEA	20	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
17	James Tate	SEA	20	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
18	James Tate	SEA	20	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
19	James Tate	SEA	20	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
20	James Tate	SEA	20	70	31	39	10	104	264	792	212	203	469	121	30	402	82	73	107	10	10	525	21	61	00
21	James Tate	SEA	20	70	3																				

Figure 1. Extract of the NBA player dataset used in the analysis (2024–25 season)

Model and Methodology

To analyze the relationship between scoring and offensive performance, a multiple linear regression model is estimated. The dependent variable of the model is points per game (PTS).

The explanatory variables included in the model are minutes played per game (MIN), field goal attempts per game (FGA), field goal percentage (FG%), three-point attempts per game (3PA), assists per game (AST), and turnovers per game (TOV). These variables were selected because they capture both offensive volume and efficiency.

The regression model is estimated using the Data Analysis Toolpak in Microsoft Excel. The aim of the model is primarily explanatory, although it also has predictive value.

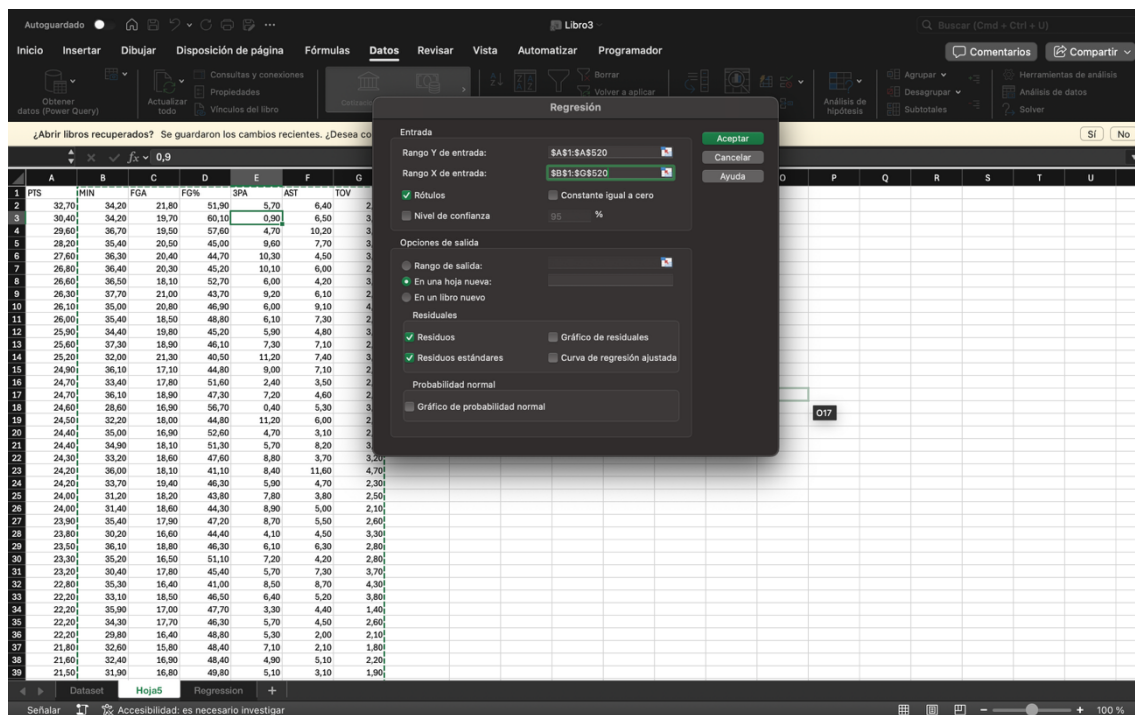


Figure 2. Configuration of the multiple linear regression model in Excel.

Results

The estimated regression model shows an excellent overall fit. The coefficient of determination is $R^2 = 0.982$, indicating that approximately 98% of the variation in points per game is explained by the independent variables included in the model.

The F-statistic is very large and statistically significant ($p < 0.001$), confirming that the model is globally significant.

1	A	B	C	D	E	F	G	H	I	J	K	L
2	Resumen											
3	Estadísticas de la regresión											
4	Coefficiente de correlación múltiple	0.99117193										
5	Coefficiente de determinación R ²	0.982421794										
6	R ² ajustado	0.982215799										
7	Error típico	0.915286625										
8	Observaciones	519										
9	ANÁLISIS DE VARIANZA											
10		Grados de libertad	Suma de cuadrados	Producto de los cuadrados	F	Valor crítico de F						
11	Regresión	6	23972.18573	3995.364288	4769.162842	0						
12	Residuos	512	428.9277979	0.837749605								
13	Total	518	24401.11353									
14												
15		Coefficientes	Error típico	Estadística t	Probabilidad	Inferior 95%	Superior 95%	Inferior 95.0%	Superior 95.0%			
16	Intercepción	-2.304866732	0.173786356	-13.26264491	1.01811E-34	-2.646288818	-1.963444647	-2.646288818	-1.963444647			
17	HNA	-0.017858483	0.01068857	-1.770169989	0.077283876	-0.037678569	0.001961603	-0.037678569	0.001961603			
18	FGA	1.393030685	0.027156528	51.26239989	6.8448E-204	1.33967922	1.44638195	1.33967922	1.44638195			
19	FG%	0.047342825	0.003956976	11.96439622	2.91628E-29	0.039568918	0.055116731	0.039568918	0.055116731			
20	3PA	-0.173200412	0.031831128	-5.444369101	8.07959E-08	-0.235836105	-0.110764719	-0.235836105	-0.110764719			
21	AST	0.026203156	0.04947103	0.602773846	0.549926532	-0.059200373	0.111604705	-0.059200373	0.111604705			
22	TOV	0.038704889	0.121967756	0.317337058	0.75117226	-0.209913954	0.278323732	-0.209913954	0.278323732			
23												
24												
25												
26												
27	Análisis de los residuos											
28		Observación	Predicción P13	Residuos	Residuos estandarizados							
29		1	29.18221406	3.517785943	3.865825021							
30		2	27.50662616	2.893373843	3.17963548							
31		3	26.5111688	3.088831202	3.394430797							
32		4	26.42782285	1.77217148	1.947510863							
33		5	26.03760239	1.562397615	1.716976498							
34		6	25.98253871	0.817461294	0.898338436							
35		7	23.94227362	2.657726377	2.920673769							
36		8	27.00266494	-0.702664943	-0.772184483							
37		9	27.63435526	-1.534355264	-1.686159723							
38		10	24.37516821	1.624831787	1.785877118							
39		11	26.02203105	-0.122031049	-0.134104831							
40		12	24.57290273	1.02709727	1.12871516							
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Figure 3. Regression summary and analysis of variance (ANOVA).

The results indicate that field goal attempts per game (FGA) have a positive and statistically significant effect on points per game. This suggests that, holding other factors constant, players who attempt more shots tend to score more points.

Field goal percentage (FG%) also shows a positive and statistically significant coefficient, highlighting the importance of shooting efficiency in scoring performance.

In contrast, three-point attempts per game (3PA) display a negative and statistically significant coefficient. This implies that a higher volume of three point attempts does not necessarily lead to higher scoring once efficiency and shot volume are controlled for.

Minutes played, assists, and turnovers are not statistically significant in the presence of the other variables, suggesting that their effects are largely captured by shooting related statistics.

	<i>Coeficientes</i>
Intercepción	-2,304866732
MIN	-0,017858483
FGA	1,393026085
FG%	0,047342825
3PA	-0,173300412
AST	0,026203166
TOV	0,038704889

Figure 4. Estimated regression coefficients and significance levels.

Model Assumptions and Diagnostics

The validity of the linear regression model was assessed through an analysis of residuals. The residuals appear to be approximately normally distributed, and no clear pattern is observed in the residuals versus fitted values plot, suggesting that heteroscedasticity is not a major concern.

Since each observation corresponds to a different player, independence between observations is assumed. Overall, the main assumptions of the linear regression model appear to be reasonably satisfied.

27	Análisis de los residuales			
28				
29	<i>Observación</i>	<i>Pronóstico PTS</i>	<i>Residuos</i>	<i>Residuos estándares</i>
30	1	29,18221406	3,517785943	3,865825021
31	2	27,50662616	2,893373843	3,17963548
32	3	26,5111688	3,088831202	3,394430797
33	4	26,42782285	1,772177148	1,947510983
34	5	26,03760239	1,562397615	1,716976498
35	6	25,98253871	0,817461294	0,898338436
36	7	23,94227362	2,657726377	2,920673768
37	8	27,00266494	-0,702664943	-0,772184483
38	9	27,63435526	-1,534355264	-1,686159723
39	10	24,37516821	1,624831787	1,785587718
40	11	26,02203105	-0,122031049	-0,134104431
41	12	24,57290273	1,02709727	1,12871516
42	13	27,10477824	-1,904778242	-2,093231227
43	14	21,7268591	3,173140904	3,487081846
44	15	24,09835485	0,601645146	0,661170093
45	16	24,575873	0,124126997	0,136407746
46	17	23,59653094	1,003469063	1,10274925
47	18	22,64401676	1,855983243	2,039608599
48	19	22,47368086	1,92631914	2,116903316
49	20	24,08459296	0,315407036	0,346612452
50	21	23,96179901	0,338200994	0,371661575
51	22	23,24193636	0,958063638	1,05285155
52	23	25,49968492	-1,299684921	-1,428271807
53	24	23,40923011	0,590769889	0,649218871
54	25	23,81187165	0,188128349	0,206741197
55	26	22,96972776	0,930272239	1,022310554
56	27	21,9171702	1,882829796	2,069111269
57	28	24,64762634	-1,147626339	-1,26116901
58	29	21,44132743	1,858672565	2,042563994
59	30	23,4441428	-0,244142796	-0,268297544
60	31	20,77275749	2,027242506	2,227811734
61	32	24,25065381	-2,05065381	-2,253539282
62	33	22,59129933	-0,39129933	-0,430013299

Figure 5. Residual analysis of the regression model.

Residual analysis is used to evaluate whether the main assumptions of the linear regression model are satisfied. In particular, it helps assess normality, homoscedasticity, linearity, and the presence of outliers.

In this study, the residuals appear to be approximately normally distributed, and no clear pattern is observed in the residuals versus fitted values plot. This suggests that the assumptions of linearity and constant variance are reasonably met.

Overall, the residual analysis supports the validity of the regression model and indicates that the estimated coefficients and statistical inferences are reliable.

In addition, although some explanatory variables are conceptually related, no severe multicollinearity issues were detected, and the estimated coefficients remain stable and interpretable.

Conclusions and Limitations

This study analyzes the relationship between offensive performance metrics and points per game in the NBA using a multiple linear regression framework. The results show that scoring is strongly associated with shot volume and shooting efficiency.

The model provides an excellent fit and highlights the key role of field goal attempts and field goal percentage in explaining scoring output. However, this analysis is subject to several limitations. The data are observational and based on per game averages, which means that causal interpretations cannot be made. Additionally, contextual factors such as team role or game situation are not included. Future research could incorporate more detailed or game level data.

